

FTC Autonomous Teacher Lesson Plan

Part 10 - Use the Built In IMU to Turn 90 Degrees

Middle and high school FTC students | 60 to 75 minutes | Teams of up to eight

Students extend the Part 9 encoder lesson by adding the REV Control Hub's built-in inertial measurement unit. They define the robot's front, document the Hub mounting orientation, verify yaw telemetry while the robot is disabled, and safely test a closed-loop 90-degree right turn.

Lesson outcome

Students can explain that encoders measure wheel travel while the IMU measures direction. They can configure the Hub's physical orientation, establish zero heading, trace a proportional turning loop, and use measured angle evidence to improve a turn.

Learning objectives

- Identify the IMU inside the Control Hub and distinguish yaw from pitch and roll.
- Mark the robot's agreed front and document the REV logo and USB-port directions.
- Match the physical Hub mounting to RevHubOrientationOnRobot parameters.
- Explain initialization, resetYaw, target heading, heading error, tolerance, timeout, and explicit stop.
- Use telemetry to verify that counterclockwise yaw is positive and clockwise yaw is negative.
- Perform a POWERING ON call-and-clear sequence before a floor-motion test.
- Measure turn error over repeated trials and change only one constant at a time.

Required materials

- [] Two-motor goBILDA StarterBot with a rigidly mounted REV Control Hub
- [] Driver Station, charged robot battery, and StarterBotImuTurn90.java
- [] Painter's tape, framing square or large protractor, and floor-safe start labels
- [] At least a six-foot-square clear test area with an additional stopping buffer
- [] Printed lesson plan, trial record sheet, and POWERING ON safety poster
- [] Tape label marked FRONT for the team-agreed front edge of the chassis

Teacher preparation

- [] Confirm the Control Hub is rigidly attached and cannot twist relative to the drivetrain.
- [] Record the actual REV logo-facing and USB-facing directions; do not assume the sample UP and FORWARD values match.
- [] Confirm hardware names are left_drive, right_drive, and imu.
- [] Build the Java file and confirm the Autonomous OpMode appears on the Driver Station.
- [] Tape a start box, a 90-degree reference line, a surrounding boundary, and a stopping buffer.
- [] Verify that the robot remains still during IMU initialization.
- [] Test at low power and keep the Driver Station operator ready to press STOP.

IMU setup reference

The sample code assumes the REV logo faces up and the USB ports face the robot's front. Change those constants when the physical Hub is mounted differently. A correct orientation lets the SDK report yaw in robot coordinates rather than Hub coordinates.

ITEM	EXAMPLE	WHAT STUDENTS VERIFY
Robot front	Tape label on chassis	The whole team agrees which edge is forward.
Configuration name	imu	The Hardware Map name exactly matches the Java string.
REV logo direction	UP	Direction the REV logo faces on the actual robot.
USB direction	FORWARD	Direction the top USB ports face on the actual robot.
Yaw at start	0 degrees	Robot is square at the start when resetYaw is called.
Right-turn target	-90 degrees	Telemetry confirms clockwise rotation decreases yaw.
Acceptable finish	-88 to -92 degrees	The heading is inside the two-degree tolerance.

Student roles

ROLE	RESPONSIBILITY
Safety captain	Controls the boundary, leads POWERING ON, and authorizes INIT and START.
Driver Station operator	Selects the OpMode and presses INIT, START, or STOP only when directed.
Robot setup technician	Marks FRONT, records Hub orientation, and squares the disabled robot.
Code reader	Finds the matching Java constants, methods, and stop conditions.
Angle observer	Reads the final angle from the floor reference without entering early.
Data recorder	Records target, final yaw, error, timeout, and trial conditions.
Two debugging analysts	Compare evidence and propose one controlled change.

Nonnegotiable safety boundary

An IMU-controlled turn must be tested on the floor because the chassis must rotate for yaw to change. No student may hold, steady, or touch the robot while POWERING ON is called or while the OpMode is enabled. Press STOP, visually confirm DISABLED, and receive the safety captain's permission before approaching.

POWERING ON for a turning robot

1. Place the robot squarely at the start while the Driver Station shows DISABLED.
2. Remove hands, tools, cables, measuring equipment, and loose objects from the turn area.
3. Check the complete sweep of the chassis and the surrounding stopping buffer.
4. Call loudly: POWERING ON!
5. Pause while everyone moves behind the safety boundary.
6. Visually confirm that no one is touching or holding the robot.
7. Safety captain authorizes INIT and START; the operator remains ready to press STOP.

Video pause guide

Use these planned pause points when the Part 10 video is available. Minor timestamp adjustments may be needed after final editing.

PAUSE	TEACHER QUESTION	EXPECTED IDEA
0:10	What information can the IMU provide that encoders cannot?	Robot direction or heading rather than wheel travel.
0:20	Where is the sensor in this lesson?	It is built into the REV Control Hub.
0:31	Which edge is our robot's front?	The team-defined edge marked FRONT.
0:43	Why must the Hub be rigidly mounted?	The sensor must rotate exactly with the chassis.
0:55	Which two mounting directions must match the code?	REV logo-facing and USB-facing directions.
1:05	Why should the robot remain still during initialization?	Movement can interfere with a reliable starting sensor state.
1:15	What does resetYaw establish?	The current aligned direction becomes zero degrees.
1:25	Which way should yaw move during a right turn?	Toward a negative angle after sign verification.
1:37	At -70 degrees with a -90-degree target, what is the error?	Negative 20 degrees; the robot must continue clockwise.
1:47	Why reduce power near the target?	To reduce overshoot and settle more accurately.
1:56	Why use a tolerance instead of demanding exactly -90.000?	Real sensors and motion vary; a small acceptable band lets the loop finish.

Teacher emphasis

Use the phrase 'encoders tell how far; the IMU tells which way.' Avoid describing the IMU as a distance sensor. Emphasize that the sample mounting directions are not universal and that telemetry must confirm the yaw sign before floor motion.

Lesson flow

STEP	ACTIVITY	TIME	TEACHER ACTION	STUDENT ACTION
1	Hook and compare sensors	5 min	Ask how the robot could know it completed a right-angle turn.	Compare timed turns, encoders, and heading feedback.
2	Watch with pauses	12 min	Use the pause guide while the robot remains disabled.	Annotate orientation, yaw, error, and stop conditions.
3	Physical setup audit	10 min	Require evidence for FRONT, logo direction, USB direction, and hardware names.	Inspect and record each setup fact.
4	Disabled telemetry check	8 min	Keep motors disabled and supervise manual chassis rotation.	Predict and verify the sign of left and right yaw.
5	Trace the turn loop	8 min	Model target minus current heading and clipped turn power.	Calculate example errors and predict motor directions.
6	First 90-degree trial	10 min	Lead the floor-motion boundary and safety sequence.	Perform assigned roles and record the result.
7	Evidence and one change	10 min	Approve one gain, speed, or tolerance adjustment.	Justify one controlled change from recorded evidence.
8	Exit explanation	5 min	Ask each student to explain one safety or control decision.	Submit an individual explanation.

Success criteria

- Code orientation matches the physical Control Hub.
- Yaw reads zero at the aligned start and changes with the expected sign.
- The turn stops inside the agreed tolerance without timing out.
- All students remain behind the boundary until STOP and DISABLED are confirmed.
- The team records evidence before changing a constant.

Hands on walkthrough

Physical orientation audit while disabled

The setup technician places the FRONT label. The group stands at the rear of the robot and agrees on robot forward, left, right, up, and down. Students then identify the direction the REV logo faces and the direction the USB ports face. The code reader finds the two matching constants in StarterBotImuTurn90.java. Every student initials the setup record.

Teacher question: If the Hub is moved later, what must be inspected before the next IMU test?

Yaw sign check with motors disabled

After INIT, confirm the Driver Station still shows the robot as disabled. One student gently rotates the entire chassis by hand while the observer watches telemetry. Rotate left, return to the marked start, rotate right, and return again. No motor command is issued. Record which direction produces positive and negative yaw.

Teacher question: Why do we verify the sign instead of memorizing a rule without checking the robot?

Trace the feedback loop

Give the group three current headings: -20, -70, and -89 degrees. For a -90-degree target, students calculate the remaining error and predict whether turn power should be large, smaller, or stopped. The code reader connects each answer to getHeadingError, Range.clip, the minimum turn power, and HEADING_TOLERANCE.

Teacher question: What could happen if maximum turn power is too high?

First floor turn

The setup technician squares the disabled robot inside the start box and leaves the area. The safety captain clears the full chassis sweep and leads POWERING ON. After approval, the operator presses INIT and START. When motion ends, the operator presses STOP and calls DISABLED. Only then may the angle observer approach and measure.

Teacher question: Did the robot finish because it entered tolerance, timed out, or was stopped by the operator?

Evidence before editing

Record target heading, final yaw, measured floor angle, elapsed time, maximum turn power, surface, and battery condition. Repeat without a code change. If results vary widely, inspect traction, loose wheels, Hub mounting, and robot setup before tuning software.

Teacher question: Which single value would you change, and what evidence supports that choice?

Technical notes

Universal IMU interface. Use `com.qualcomm.robotcore.hardware.IMU` with `RevHubOrientationOnRobot`. Avoid beginning a new lesson with the legacy BNO055IMU interface.

Robot coordinates. Yaw is rotation around the robot's vertical Z axis. Under the right-hand rule, counterclockwise is positive and clockwise is negative when viewed from above.

Initialization. Initialize the IMU during setup and keep the robot still. Call `resetYaw` only after the chassis is aligned with the intended zero direction.

Normalized error. `AngleUnit.normalizeDegrees` keeps heading error between -180 and +180 degrees so the robot chooses the shorter turn across the wrap point.

Proportional turning. Turn power is heading error multiplied by `TURN_GAIN`, then clipped to `MAX_TURN_POWER`. A small minimum power can overcome static friction outside tolerance.

Tolerance. A two-degree band is an instructional starting point. Tight tolerances can cause oscillation or prevent completion.

Independent stop paths. Target tolerance, timeout, Driver Station STOP, and the finally block must all result in zero motor power.

Floor testing. A wheels-up test can verify motor direction, but it cannot validate a chassis heading change because the robot itself does not rotate.

Assessment evidence

- Accurate physical Hub-orientation record
- Correct explanation of yaw and `resetYaw`
- Correct heading-error calculations
- Unprompted completion of the call-and-clear sequence
- Two or more controlled turn records
- Evidence-supported recommendation for one code change

Exit prompt

In two or three sentences, explain how the program knows whether to keep turning, slow down, or stop. Include the words target heading, current heading, error, and tolerance.